Hazus 7.1 Release Notes

This guide provides a summary of the Hazus 7.1 release, including new features and enhancements as well as resolved issues.

Introduction

Welcome to the Release Notes for Hazus 7.1, FEMA's natural hazard risk assessment software. This document provides a comprehensive overview of the latest enhancements, performance improvements, and user experience updates included in this release. In addition to outlining new features, it covers system requirements, known limitations, and a preview of future developments in the Hazus program.

What to Expect in this Guide

- New Features and Enhancements
- Resolved Issues in Hurricane Model
- Known Software Limitations
- Additional Resources
- Future Updates

Release Date: Sep. 3, 2025

What's New

Hazus 7.1 represents a significant update in natural hazard risk assessment modeling. For more details on the improvements made in this release and our future plans, please continue reading the full document. Below are some of the key highlights introduced in Hazus 7.1:

- Upgrade in Place: Users will not have to uninstall, risk losing data or start over with the latest release. Hazus 7.1 is available as a patch on top of Hazus 7.0.
- User Supplied Hurricane Scenarios: Following National Hurricane Center data import functionality in Hazus 7.0, users can now import custom windspeed data.
- Results Export: Users can export analysis results right from the Hazus workflow, streamlining a process that was previously only available as an open-source Hazus Export tool.
- Hurricane Fixes: Updates to hurricane damage functions and Specific Building Type (SBT) damage state building count calculations, and a repackaging of hurricane state data, are available.
- Elimination of Security Vulnerabilities: All levels of security vulnerabilities identified through extensive scanning since the last release have been addressed to ensure a secure user experience.

System Requirements

For optimal performance of Hazus 7.1, it is recommended to adhere to the following system specifications. While the software may function on lower-spec systems, doing so could result in slower performance and potential instability.

Table 1: System and Software Requirements for Hazus 7.1

Component	Minimum Requirements	Recommended Requirements
Operating System (OS)	Microsoft Windows v10 & 11 (Pro & Enterprise) Please Note: other OS have not been fully tested and may not function as expected.	Latest version of Microsoft Windows v11 (Pro & Enterprise)
Windows .NET Framework Version	8*	-
Processor	2.4 GHz with at least 2 cores and simultaneous multithreading	4 cores (or higher)
Memory/RAM	8 GB	16 GB or more
Hard Drive	50 GB of free space	100 GB or more of free space for large study areas
Screen Resolution	1024 x 768 pixels	1920 x 1080 pixels (or higher)
Video/Graphics Adapter	24-bit video card with a minimum of 128 MB video memory	-
ArcGIS Pro Version, License Type, and Extensions	ArcGIS Pro v3.4 (Basic, Standard, or Advanced); Spatial Analyst extension is required to run a flood analysis	ArcGIS Pro v3.4 (Standard or Advanced); Spatial Analyst extension is required to run a flood analysis. Basic licenses may experience some limitations.
Other Software Dependencies	SQL Server Express 2022	-

*On Nov. 12, 2024, Microsoft ended support for .NET 6. ArcGIS Pro versions 3.0 through 3.2 require .NET 6 to run, and this requirement will not change. For organizations unable to continue using .NET 6 due to Microsoft's end-of-life designation, Esri offers versions of ArcGIS Pro built on .NET 8 beginning with ArcGIS Pro 3.3 released in May 2024 and continuing with ArcGIS Pro 3.4 released in November 2024.

Please Note: Users must install ArcGIS Pro and Hazus 7.0 first. If help is required getting ArcGIS Pro installed, please check out Esri’s help page or reach out to their help desk for assistance. The SQL Server Express 2022 installation will be handled automatically by the Hazus 7.0 installer.

Documentation

To assist users in getting the most out of Hazus 7.1, the following resources are available:

- Hazus 7-1 Getting Started Guide: This guide offers essential information for new users, including installation steps, basic usage, and initial setup guidance.

Installation and Uninstallation

Step-by-step instructions for installing Hazus 7.1 are included in the Hazus 7-1 Getting Started Guide. Step-by-step instructions for uninstalling Hazus 7.1 are included in the Hazus 7-1 Uninstallation Guide. Both guides are available with the Hazus patch download.

New Features and Enhancements

Upgrade in Place

This feature allows new functionality to be delivered incrementally as it becomes available. Users will not have to uninstall, risk losing data or start over with the latest release. Hazus 7.1 is available as a patch on top of Hazus 7.0. Users must have ArcGIS Pro and Hazus 7.0 installed first.

The Hazus 7-1 Getting Started Guide walks through the Installation Instructions in detail.

Figure 1. Hazus 7.1 Patch Installation

User Supplied Hurricane Scenarios

Following National Hurricane Center data import functionality in Hazus 7.0, users can now import custom windspeed data.

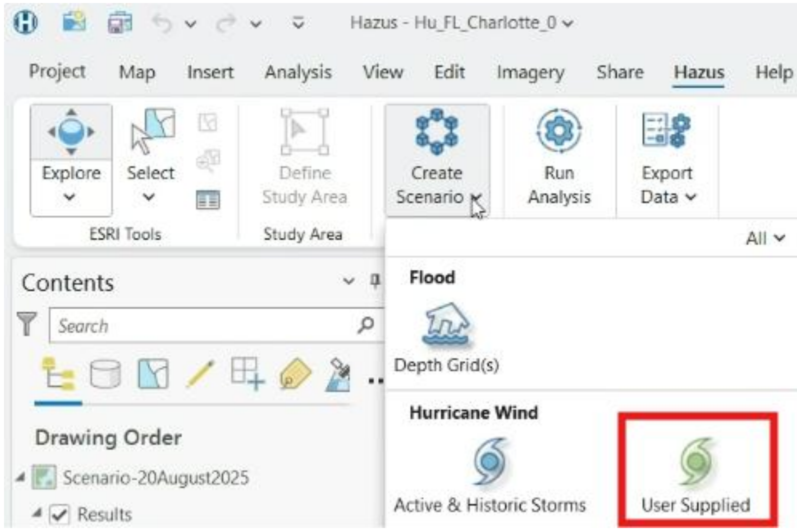

Figure 2. Create User Supplied Scenario

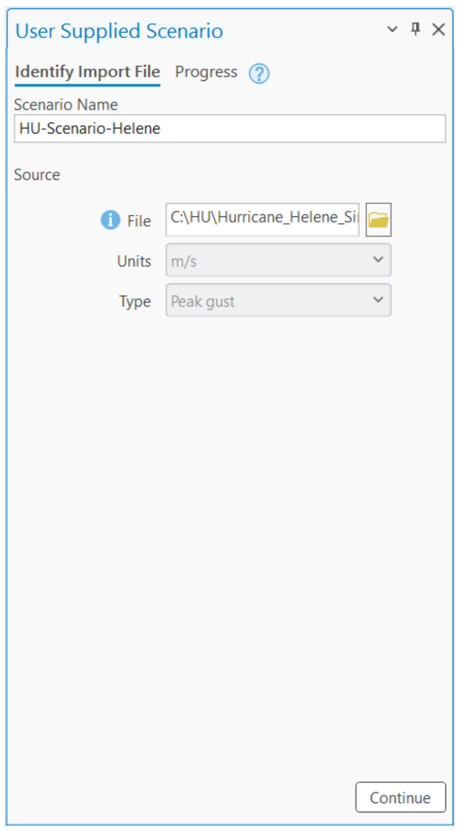

Figure 3. User Supplied Scenario Data

Users continue with the existing Hazus workflow for running the analysis and reviewing results.

Results Export

Users can export analysis results right from the Hazus workflow, streamlining a process that was previously only available as an open-source Hazus Export tool.

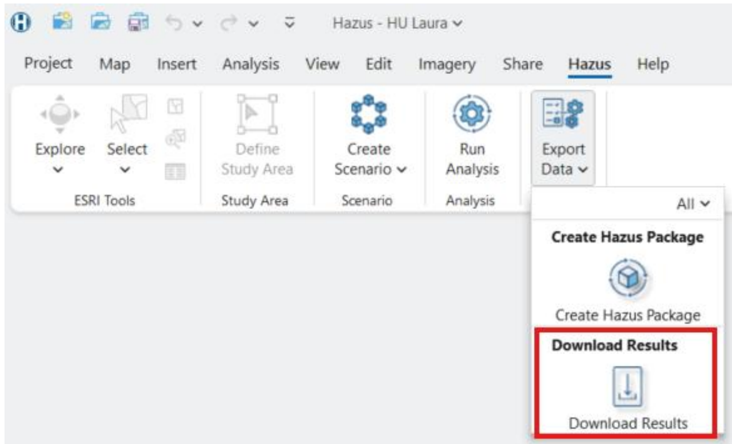

Figure 4. Download Results Selection

The Download Results tool produces a variety of results files (including .csv, .shp, GeoJSON, .png and .pdf formats), including a hazard-specific summary report and the opportunity to email others a summary of the Hazus analysis. Please note that the Flood version of the summary report is not yet available.

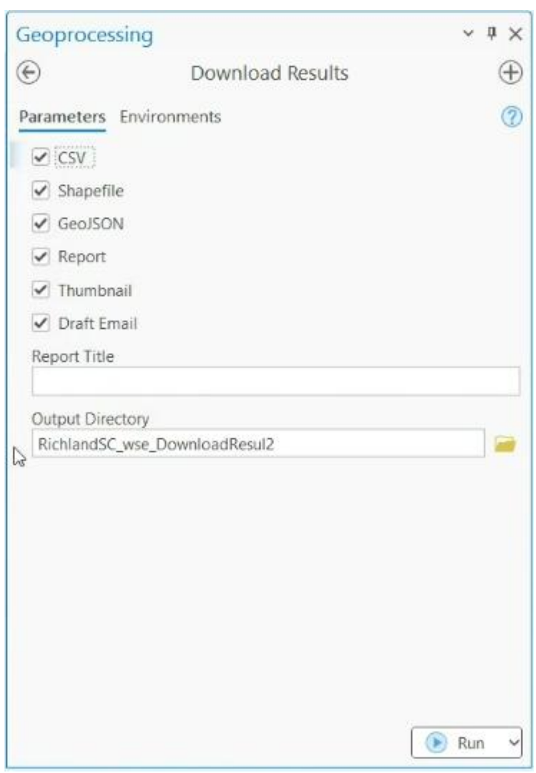

Figure 5. Download Results Setup

Resolved Issues in Hurricane Model

The Hazus 7.1 release addresses defects and issues from previous versions, improving stability and performance. Below is a list of key issues that have been resolved to enhance the overall user experience.

RSA-21186: Errors in Wind Damage Functions Specific Building Type

Background: The Hazus Hurricane Wind Model currently contains over 275,000 damage functions to represent a broad range of building types and construction details throughout hurricane states on the Gulf and Atlantic Coast, as well as Hawaii and the Caribbean. The Natural Hazards Engineering Research Institute (NHERI) SimCenter identified an issue where damage state 4 (destruction) was more likely than the other damage states for some types of 3-story multi-family residential buildings.

While addressing the issue described above, the team also discovered that the 2- and 3-story multi-family buildings were using the same damage functions as the 1-story multi-family buildings.

Finally, it was also determined that the Specific Building Type (SBT) damage state building counts were being computed incorrectly.

How it was Addressed: The issue of incorrect damage state functions for the 2- and 3-story multi-family residential buildings was addressed in two steps. First, the team was able to retrieve the correct damage state functions for 2- and 3-story multi-family buildings that were included in the original 2003 release of the Hazus hurricane model. These damage functions had been inadvertently overwritten with a copy of the 1-story multi-family building damage functions in a subsequent release. This update corrected the damage state functions for 2- and 3-story multi-unit housing that do not have shutters and have a roof deck attachment of either 6d nails @ 6"/12" spacing (rda6d) or 8d nails @ 6"/12" spacing (rda8d). However, the team still did not have the correct damage functions for mitigated 2- and 3-story multi-unit housing, as these cases were not included in the original version of Hazus. The mitigation cases included at least one of: shutters (shtys), 6d/8d mix @ 6"/6" (rda6s), and 8d @ 6"/6" (rda8s) roof deck attachment.

To develop replacement damage functions for the missing cases, the team used the damage functions for the most similar building with the same number of stories (i.e., 2- or 3-story buildings) without mitigation features as a starting point and adjusted them until the average per storm damage state matched the values given in Tables 5-57 through 5-59 in version 5.1 of the Hazus Hurricane Technical Manual.

To address the issue of the Specific Building Type damage state building counts being computed incorrectly, the calculation was corrected such that the Specific Building Type building counts were multiplied by the Specific Building Type probability for a damage state (rather than the General Building Type (GBT) probability for a damage state).

RSA-39732: Repackaged State Data

Background: The Hazus Hurricane Wind Model would encounter an out of memory error for large study areas in Hazus 6.1 and 7.0. For example, this error was likely to be encountered when attempting to run an analysis on a study area comprising the entire state of Florida, Texas, or New York. The out of memory error occurred because the

General Building Type (GBT) mapping schemes changed from one per state in version 6.0 to one per census tract in versions 6.1 and 7.0.

How it was Addressed: To reduce the memory demand, the General Building Type (GBT) mapping schemes in the hurricane model were rolled up to the same regional scales as used for the Specific Building Type (SBT) and Wind Building Characteristic (WBC) mapping schemes. For example, there are seven regions in Florida, and two in most other states (coastal and inland).

Following this change, state/region data was repackaged and made available for download through the FEMA Map Service Center (MSC) for Hazus 6.1 users. The latest state data is managed automatically for Hazus 7.1 users, through functionality introduced in Hazus 7.0, for new study areas.

The impact of this change is that the General Building Type distributions are now less precise. This simplification is most likely to impact the damage and loss estimates for urban and commercial areas, where the mix of masonry, wood frame, steel, reinforced concrete, and manufactured housing buildings is most likely to differ from more suburban and rural areas. Users with existing hurricane analyses may find differences if they aggregate identical Hazus 7.1 study areas with the latest data, due to modifications of the hurricane mapping schemes.

Known Software Limitations

While Hazus 7.1 brings significant improvements as described above, the limitations described in the Hazus 7.0 release notes still apply. They are repeated here for convenience.

Earthquake and Tsunami Models

At present, Hazus 7.1 does not support earthquake and tsunami models, which were available in previous versions supported by ArcMap. These models are slated for reintroduction in future releases, ensuring that users will eventually have the same comprehensive capabilities for earthquake and tsunami risk assessments as with other hazards such as floods and hurricanes. Users needing to run earthquake or tsunami analyses should continue to use Hazus 6.1 in the meantime. Users are also encouraged to check out the Hazus Loss Library for previously run earthquake and tsunami Hazus analyses.

Low-End Systems

While Hazus 7.1 is designed to run efficiently on modern hardware, users operating on systems that only meet the minimum specifications (8 GB of RAM, dual-core processor, etc.) may experience slower performance, particularly when running large-scale analyses. For the best user experience, users should upgrade to at least 32 GB of RAM and a quad-core processor to optimize the speed and efficiency of the software.

Windows Only (No Mac or Linux OS Support)

Hazus 7.1 is only supported on Windows machines, and currently cannot run on Mac or Linux operating systems. Users with non-Windows machines can still run Hazus using virtualization tools, however success is not always guaranteed.

Maximum 10GB Data for Master Database Files

Hazus 7.1 imposes a 10 gigabyte (GB) limit on the size of Master Database Files (MDFs) used within the software. For larger datasets, users may need to split their data into smaller, more manageable files or compress them. Exceeding this limit can cause performance issues or errors during analysis.

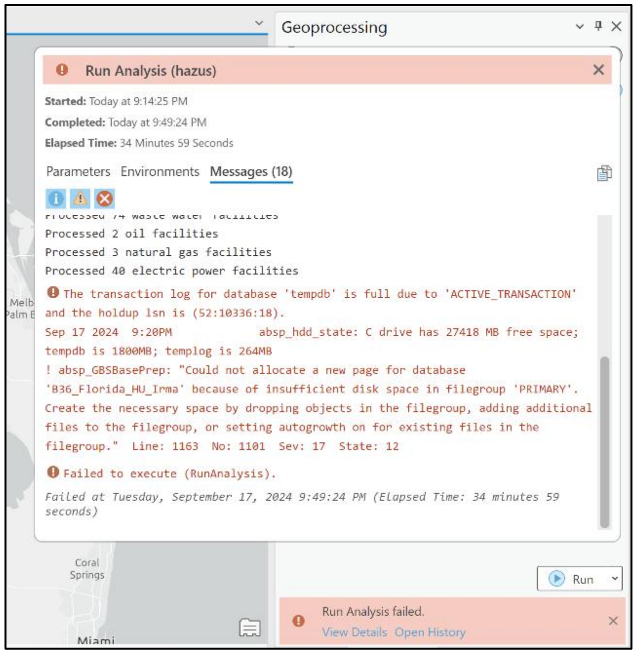

Figure 6. Error message when database exceeds the size limit

To resolve this issue, users will need to break their study area into smaller study areas and re-run the analysis so that it does not exceed the database size limitation.

HTML Help Pages Will Not Open If Username Contains Spaces

If the user’s Windows username contains spaces, they will not be able to open the HTML help pages in Hazus 7.1.

Pipeline Inventory Data May Appear Outside Study Area

When defining study areas that include pipeline inventory data or other line feature datasets, users may notice that these features sometimes appear outside the study area boundary. This occurs because, by design, the software does not clip or mask line features to the study area; even a small overlapping section causes the entire line to be included.

Additional Resources

Training and Support

Hazus offers a range of resources to help users get the most out of the software:

- Documentation and Guides: Explore the Hazus 7-1 Getting Started Guide and other Hazus Documentation for comprehensive instructions on installation, setup, and advanced workflows.

Community and Feedback

Stay connected and share insights with the Hazus team:

- GovDelivery Email List: Stay updated on new releases, feature updates, and upcoming training opportunities by joining our GovDelivery email list.
- Provide Feedback: Feedback is crucial for improving Hazus. Share suggestions, experiences, and any issues encountered with the Hazus Help Desk to help make Hazus better.

Support Contacts

If assistance is needed, the Hazus support team is here to help:

- Hazus Help Desk: For any questions related to this release, technical support, troubleshooting, or general guidance, contact the Hazus Help Desk.

Future Updates

The Hazus team is committed to continuing to improve the software and working toward a release schedule of two updates per year to introduce new features and enhancements. While this is our goal, it is important to note that the schedule may vary depending on development progress and user needs.

For the Hazus desktop version, the primary goal continues to be reintroducing the earthquake and tsunami modeling capabilities, ensuring Hazus once again supports all four major hazards: flood, hurricane, earthquake, and tsunami. Users can also expect improvements to performance, user workflows, and overall software functionality, based on community feedback.

Looking further ahead, research into a web-based environment continues to be explored through OpenHazus, with a vision of streamlining workflows and improving accessibility to quality, easy-to-use risk assessment information.

Stay tuned for more detailed announcements on these exciting future developments!